PRODUCT USER MANUAL

For Level 4 ODYSSEA Sea Surface Temperature multi-year Product over the European North West Shelf / Iberia Biscay Irish areas

SST_ATL_L4_REP_OBSERVATIONS_010_026
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Contributors : Emmanuelle Autret (Ifremer), Jean-François Piollé (Ifremer)

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MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

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a) Structure and semantic of NetCDF maps files..... **Erreur ! Signet non défini.**

GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the Product User Manual describing the Copernicus Marine ODYSSEA ATL reprocessed level 4 product the European North West Shelf/Iberia Biscay Irish areas, which was processed at Ifremer (FR). The manual covers: how it was created, its content, and which data services are available to access it.
The Copernicus Marine ODYSSEA reprocessed L4 product provides a regional gap-free daily Sea Surface Temperature (SST) from multiple infra-red instruments onboard polar satellites. The product provides an estimate of the foundation SST which is the SST free of diurnal variability, created by the Ifremer ODYSSEA system using re-processed ESA SST CCI, C3S [RD.1]. The product is available from 1st January 1982 to 1 month before real time on a regular grid spanning from 21° W to 13° E and 9° N to 62° N at 0.05° resolution. The product is published on the Copernicus Marine dissemination server after automatic and human quality controls. Files downloaded are in NetCDF format and follow CF-1.7 and GHRSST format conventions.
The analysis system is described in the Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SST-ATL-SST-REP-OBSERVATION-010-026
Geographical coverage	European North West Shelf / Iberia Biscay Irish areas : 21°W-13°E ; 9°N – 62°N
Variables	• Analysed_sst • Analysis_error
Product Type	Reprocessing
Available time series	01/01/1982 – 1 month before real time
Temporal resolution	Daily
Update Frequency	1 month
Target delivery time	Monthly, the 25th, at 12 pm (noon) UTC
Horizontal resolution	1/20 degree
Number of vertical levels	Surface only
Format	NetCDF4 (CF1.7, GDS 2.1 conventions)

b) Production System Description

The Copernicus Marine ODYSSEA reprocessed L4 product provides a regional gap-free daily Sea Surface Temperature fields built from the L3S products (SST-ATL-PHY-l3S-MY-010-038). The satellite derived SST data which are used to reconstruct the daily SST fields, are extracted from the global ESA Climate Change Initiative (CCI) Copernicus Climate Change Service (C3S) dataset. (Embury et al., 2024 [R.1]). This archive includes observations from satellite thermal infra-red measurements from Advanced Very High Resolution Radiometers (AVHRRs), Along-Track Scanning Radiometers (ATSRs), the Sea and Land Surface Temperature Radiometer (SLSTR) on-board SENTINEL-3 and from microwave observations as described in [R.1]. Only observations with quality level values greater than 4 (out of 5, corresponding to acceptable and best quality measurements in GHRSST format convention) have been included in the analysis. In order to avoid the use of data affected by diurnal warming, only night-time observations have been included.

L3S : daily adjusted sea surface temperature

The L3S product is derived from a collection of sea surface temperature (SST) data acquired by various instruments, resulting in multiple SST observations for each pixel on a given day. Due to differences in the spectral channels used and variations in instrument sensitivity, SST measurements from different sensors may exhibit slight discrepancies. These differences typically occur at a spatial scale of approximately 100 km. Directly aggregating data from these diverse sources without accounting for such biases could introduce artifacts, including discontinuities and noise, into the L3S SST products.

To ensure the consistency of the daily L3S dataset, intercalibration of the various data sources is required. This process involves estimating a daily correction to be applied to SST values from each individual satellite source. First, a daily reference SST field is constructed by computing the median of all available observations. Then, for each satellite data source, a large-scale bias field is derived by calculating the difference between the observations and the daily reference, with the anomaly smoothed using a 70 km Gaussian filter. The estimated correction is subsequently subtracted from the SST values of each satellite source, producing the L3S *adjusted_sea_surface_temperature* variable.

The daily individual sensor composite files are then merged into a combined multi-sensor composite file (L3S) at the final product resolution. The merging process takes into account a hierarchy of sensors (from best to worst) (again, estimated from comparisons of each source with in situ measurements). Any given cell of the final grid will contain a value from one source only, which is means the best is selected. The used source is reported in the *sources_of_sst* variable of the created L3S.

L4 : daily analysed SST and analysis error

The reconstruction of the daily gap-free SST time series relies on the interpolation at each point of the time series with a Kalman smoother. This interpolation has been carried out

on the time series of SST anomalies obtained by removing a first-guess SST. The method is detailed in the Quality Information Document (QUID).

The ODYSSEA processing chain was developed and is maintained by the Laboratory for Ocean Physics and Satellite remote sensing (LOPS) at the Institut Français de Recherche pour l'Exploitation de la Mer (Ifremer, http://www.ifremer.fr/). It is run operationally for CMEMS SST TAC production at Ifremer marine data operational service (*SISMER* : Systèmes d'informations scientifiques pour la Mer).

ATL (21°W-13°E; 9°N – 62°N): 1/20°, regular grid, 680 x 1062 x 1
with 1time step per file
SST provided over European North West Shelf/Iberia Biscay Irish areas

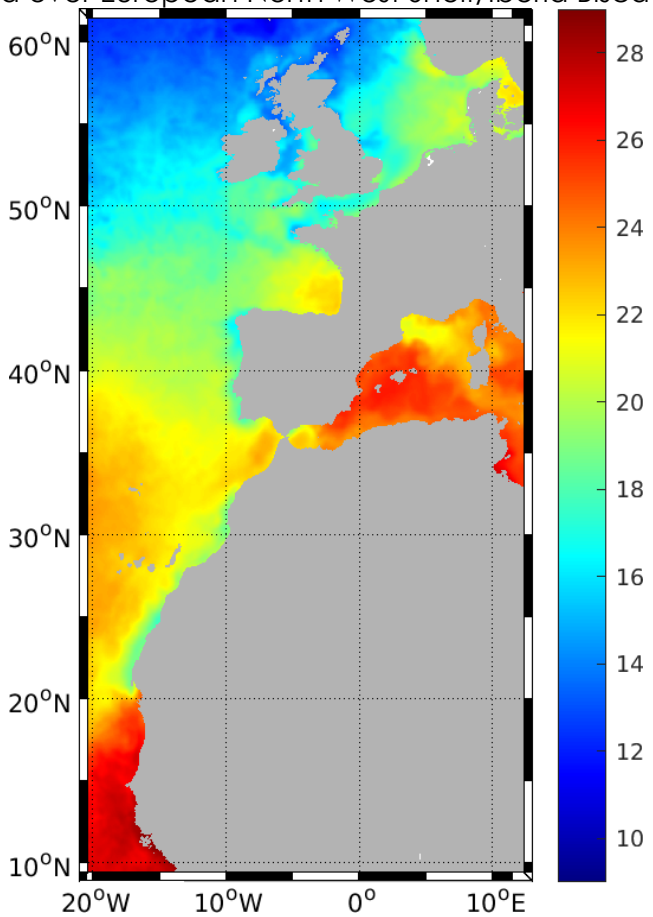

Figure 1 Illustration of the rea covered by the ATL product. The map shows an example of the reconstructed SST field (from the SST_ATL_L4_REP_OBSERVATIONS_010_026 product, on 29/07/2014)

Vertical grid	Surface only (20 cm)
System Name	ODYSSEA-REP version 3.0
Horizontal Resolution	1/20°
Input observations	SST satellite data: A collection of L3U and L3C data from various sources (ESA SST CCI, C3S and EUMETSAT; see quality information document for full details). All data are provided with Sensor Specific Error Statistics (SSES_bias and SSES_std variables) that give an estimate of the systematic

	and random errors at pixel level, and quality level flags. A bias correction is performed using ESA SST CCI / C3S data [RD.1] as reference.

c) Dataset details

Product Lines	SST_ATL_SST-REP-OBSERVATION_010_026	
Geographical coverage	ATL	
cmems-IFREMER-ATL-SST-L4-REP-OBS_FULL_TIME_SERIE		
Product Type	Multi-year – L4	
Update frequency	Montly	
Available time series	01/01/1982 – 1 M	
Temporal resolution	Daily mean	

Dataset : cmems-IFREMER-ATL-SST-L4-REP-OBS_FULL_TIME_SERIE contains daily gap-free 2D fields of foundation sea surface temperature.
analysed_sst [K] analysed sea surface temperature sea_surface_foundation_temperature
analysis_error [K] estimated error standard deviation of analysed_sst sea_surface_foundation_temperature_standard_error

d) Additional information on some parameters

N/A

e) Processing information

Product disseminating schedule :

Type of product	Observation (base time)	Product temporal coverage (hours)	Product disseminating scheduled time
Reprocessing	12 UTC	-12 to +12	Monthly

f) Catalogue updating

The catalogue data cover 1982 to present minus one month. It is updated monthly.

g) Time averaging

Dataset frequency	Time averaging
daily	daily means over a day (centered at noon).

h) Product use and impact

This product is meant for a large range of applications requiring foundation (not affected by diurnal cycle) high resolution sea surface temperature at global scale. The L3S provides the satellite observations, contrary to Level 4 datasets whose gap free sea surface temperature comes at the cost of optimal interpolation which tends to smooth the observations and is prone to errors in the estimation of the sea surface temperature over areas with persistent cloud coverage. As a consequence, the L3S provides higher and sharper resolution of ocean features. Users that want to rely on true observations rather than interpolated observations should use the L3S.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Example :

20230101000.nc000-IFR-L3S_GHRSST-SSTfnd-ODYSSEA-ATL_MY_005_adjusted-v03.0-f01.0.nc

4.2 FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [MB]
cmems-IFREMER-ATL-SST-L4-REP-OBS_FULL_TIME_SERIE	<date>000000-IFR-L4_GHRSST-SSTfnd-ODYSSEA-ATL_MY_005-v03.0-fv01.0.nc	0.7

4.3 STRUCTURE AND SEMANTIC OF NetCDF MAP FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-026-038-039-044.pdf

[R.1] Embury, O., Merchant, C. J., Good, S. A., Rayner, N. A., Høyer, J. L., Atkinson, C., ... & Donlon, C. (2024). Satellite-based time-series of sea-surface temperature since 1980 for climate applications. *Scientific Data*, *11*(1), 326.

6) ANNEX

An example header of a high resolution L4 netCDF file (generated using ncdump) is given below.

```
netcdf \20140729000000-IFR-L4_GHRSST-SSTfnd-ODYSSEA-ATL_005-v03.0-fv1.0 {
dimensions:
        time = 1 ;
        lat = 1062 ;
        lon = 680 ;
variables:
        double time(time) ;
                time:long_name = "reference time of sst field" ;
                time:standard_name = "time" ;
                time:axis = "T" ;
                time:units = "seconds since 1981-01-01 00:00:00" ;
                time:calendar = "Gregorian" ;
        double lat(lat) ;
                lat:long_name = "latitude" ;
                lat:standard_name = "latitude" ;
                lat:axis = "Y" ;
                lat:units = "degrees_north" ;
        double lon(lon) ;
                lon:long_name = "longitude" ;
                lon:standard_name = "longitude" ;
                lon:axis = "X" ;
                lon:units = "degrees_east" ;
        short analysed_sst(time, lat, lon) ;
                analysed_sst:_FillValue = -32768s ;
                analysed_sst:long_name = "analysed sea surface temperature" ;
                analysed_sst:standard_name = "sea_surface_foundation_temperature" ;
                analysed_sst:type = "foundation" ;
                analysed_sst:units = "kelvin" ;
                analysed_sst:add_offset = 273.15 ;
                analysed_sst:scale_factor = 0.01 ;
                analysed_sst:valid_min = -300s ;
                analysed_sst:valid_max = 4500s ;
        short analysis_error(time, lat, lon) ;
                analysis_error:_FillValue = -32768s ;
```

```
                    analysis_error:long_name = "estimated error standard deviation of
analysed_sst" ;
                    analysis_error:type = "foundation" ;
                    analysis_error:units = "kelvin" ;
                    analysis_error:add_offset = 0. ;
                    analysis_error:scale_factor = 0.01 ;
                    analysis_error:valid_min = 0s ;
                    analysis_error:valid_max = 32767s ;
// global attributes:
                    :Conventions = "CF-1.4" ;
                    :title = "European North West Shelf/Iberia Biscay Irish Seas - High
Resolution L4 Sea Surface Temperature Reprocessed" ;
                    :summary = "A merged, multi-sensor L4 Foundation SST product"  ;
                    :id = "IFR-L4_GHRSST-SST-ODYSSEA-ATL" ;
                    :cmems_product_id =
"SST_ATL_SST_L4_REP_OBSERVATIONS_010_026" ;
                    :references = "" ;
                    :institution = "Ifremer" ;
                    :license = "CMEMS data use is free and open. Users must register at
CMEMS help desk: servicedesk.cmems@mercator-ocean.eu" ;
                    :naming_authority = "org.ghrsst" ;
                    :uuid = "" ;
                    :contact = "emmanuelle.autret@ifremer.fr,jean.francois.piolle@ifremer.fr" ;
                    :gds_version_id = "2.0" ;
                    :product_version = "2.0" ;
                    :netcdf_version_id = "4.1.1 of Apr  7 2011 12:37:05" ;
                    :date_created = "20200828T172256Z" ;
                    :history = "analysis originally produced by Ifremer/Cersat with Odyssea
processor 3.0" ;
                    :platform = Sential-<3a,3b>, ERS<1,2>, NOAA-
<06,07,08,09,10,11,12,14,15,16,17,18,19>, MetOp,NOAA-20" ;
                    :sensor = "SLSTRA, ATSR1,
ATSR2,AATSR,AVHRR<06,07,08,09,10,11,12,14,15,16,17,18,19>, AVHRRMTA" ;
                    :source_data = "SLSTRA-C3S-L3C-v2.0, ATSR<1,2>-ESACCI-L3U-v2.0,
AATSR-ESACCI-L3U-v2.0, AVHRR<06,07,08,09,10,11,12,14,15,16,17,18,19>-ESACCI-L3U-
v2.0, AVHRRMTA-ESACCI-L3U-v2.0," ;
                    :processing_level = "L4" ;
                    :cdm_data_type = "grid" ;
                    :spatial_resolution = "0.05 degree" ;
                    :temporal_resolution = "daily" ;
                    :start_time = "20140728T120000Z" ;
                    :stop_time = "20140729T120000Z" ;
                    :southernmost_latitude = "8.92" ;
                    :northernmost_latitude = "61.98" ;
                    :westernmost_longitude = "-20.98" ;
                    :easternmost_longitude = "12.98" ;
                    :geospatial_lat_units = "degrees_north" ;
                    :geospatial_lat_resolution = "0.05" ;
                    :geospatial_lon_units = "degrees_east" ;
                    :geospatial_lon_resolution = "0.05" ;
                :file_quality_index = "3" ;
```